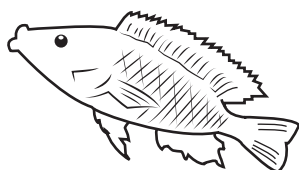
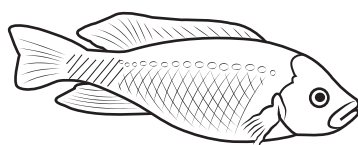
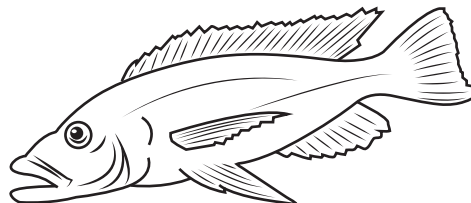


- 68 Many different species of cichlid fish have evolved from a common ancestor. The drawing shows some of the species that have evolved in Lake Victoria in Africa.

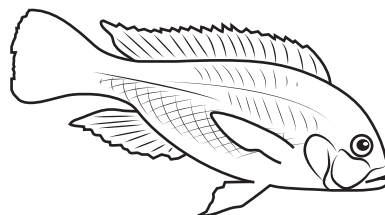
Haplochromis chilotes
(feeds on insects)



Haplochromis macrognathus
(feeds on other fish)



Astatotilapia elegans
(generalised bottom feeder)



Macropodus bicolor
(feeds on snails and other molluscs)

Which of the following statements about evolution of the cichlid fish is/are correct?

- 1 Different food types being available caused mutations in the DNA of the ancestral population of fish.
 - 2 Natural selection was possible because there were multiple alleles of the genes affecting the traits shown in the diagram.
 - 3 Natural selection of individuals with phenotypes best suited to different parts of the environment took place.
- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3